

A negative proposed Jarník dimension

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Conjecture 00000008341

Result. The stated dimension formula $2 - (w + 1)$ is false at $w = 2$: it predicts -1 for a nonempty subset of the real line.

The approximation set. Use the standard Jarník parameter convention in which the error is $q^{-(w+1)}$. For integer $w \geq 1$, define

$$W(w) = \left\{ x \in [0, 1] \setminus \mathbb{Q} : \begin{array}{l} \text{for every } Q \in \mathbb{N} \text{ there are } q \in \mathbb{N}, p \in \mathbb{Z}, \\ q > Q, \quad 0 < |x - p/q| < q^{-(w+1)} \end{array} \right\}.$$

Thus the denominators are genuinely unbounded. Restricting to irrationals removes exact rational coincidences. This is the classical well-approximable set underlying the Jarník–Besicovitch spectrum; see [1], Sections 2.3 and 3.2.

An explicit point. Let $L = \sum_{n=0}^{\infty} 10^{-n!}$. Its first two terms total $1/5$ and its remaining tail is strictly between 0 and $1/10$, so $0 < L < 1$. The factorial truncations give the Liouville approximation property: for every integer exponent N there are strict rational approximations with error less than q^{-N} . Such approximations have arbitrarily large denominators. Otherwise, irrationality gives a positive minimum distance from L to fractions with the finitely many bounded denominators in a fixed bounded interval; taking N sufficiently large contradicts that minimum. Consequently $L \in W(w)$ for every integer w , in particular $W(2) \neq \emptyset$.

Dimension and contradiction. The actual Hausdorff dimension is monotone under inclusion, and the real line has Hausdorff dimension one. Hence

$$0 \leq \dim_H W(2) \leq \dim_H \mathbb{R} = 1.$$

It is therefore finite and cannot equal $2 - (2 + 1) = -1$. Both source languages explicitly print subtraction in $2 - (w + 1)$, with no restriction excluding $w = 2$. This disproves the stated dimension conjunct, hence the combined conjecture. No assertion concerning its other uniform/nonuniform or convexity clauses is required. The classical formula is $2/(w + 1)$ for $w > 1$ [1]; that deeper theorem is not needed for this elementary contradiction.

Formal verification. Lean defines the full quantified set above, verifies the actual factorial-series witness using Mathlib’s Liouville theorems, and proves its interval bounds. It uses Mathlib’s genuine `MeasureTheory.dimH`, establishes the bound by one and finiteness, then takes `ENNReal.toReal` to compare the dimension with the real number -1 . No negative number is cast into extended nonnegative reals. The final theorem `Counterexample.counterexample` includes nonemptiness, finiteness, both dimension bounds, and failure of the proposed equality.

References

- [1] V. Beresnevich, F. Ramírez and S. Velani, *Metric Diophantine Approximation: aspects of recent work*, in *Dynamics and Analytic Number Theory* (2017), Sections 2.3 and 3.2, Theorem 3.2. Author manuscript.